

# The Exact Uniform Scrambling Horizon of the Order-Four Almost-Sarymsakov Class Is Eleven

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China

xiaojianzeng@foxmail.com

## Abstract

For one fixed common ordering, let  $\mathcal{A}_4$  be the full order-four almost-Sarymsakov class, namely the union of its four admissible mixing-core levels. We prove that its exact uniform scrambling horizon is

$$h_{\text{scr}}(\mathcal{A}_4) = 11.$$

The lower bound is an explicit ten-factor word of legal Boolean supports. The upper bound is a finite, exhaustive computation over all  $15^4 = 50,625$  row-nonempty 4-by-4 supports: the complete nonscrambling frontier has 120 states after ten factors and is empty after eleven. A separately implemented checker reconstructs the legal alphabet and every frontier state. Because support multiplication is exact for nonnegative matrices, the conclusion covers arbitrary positive weights compatible with every admitted support, not only sampled matrices.

## 1 Statement and context

We use the definitions and the single fixed ordering of Hsu [1, Definition 3]. The notation is recalled here so that the claim and its computational reduction are unambiguous.

Let  $\mathcal{S}_4$  denote the row-stochastic matrices of order four, and let  $P = (p_{ij}) \in \mathcal{S}_4$ . For a row-allowable square nonnegative matrix  $B$  of order  $s$ , write

$$F_B(I) = \{j \in [s] : b_{ij} > 0 \text{ for some } i \in I\}, \quad I \subseteq [s].$$

The matrix  $B$  is *GE* when, for every pair of disjoint nonempty  $I, J \subseteq [s]$ , either  $F_B(I)$  and  $F_B(J)$  intersect or

$$|F_B(I) \cup F_B(J)| \geq |I \cup J|.$$

It is *GR* when the displayed inequality is strict whenever those two image sets are disjoint. Row-allowability means that every row contains a positive entry; for a matrix with no rows this condition is vacuous.

Let  $P^{[s]}$  be the leading  $s$ -by- $s$  block of  $P$ , and let  $P_{21}^{[r]}$  be the block in rows  $r+1, \dots, 4$  and columns  $1, \dots, r$ . Define

$$\begin{aligned} \mathcal{T}_{\text{NE},4}(r) &= \{P \in \mathcal{S}_4 : P^{[r]} \text{ is GR and } P^{[s]} \text{ is GE for } r < s \leq 4\}, \\ \mathcal{T}_{\text{AS},4}(r) &= \{P \in \mathcal{T}_{\text{NE},4}(r) : P_{21}^{[r]} \text{ is row-allowable}\}, \\ \mathcal{A}_4 &= \bigcup_{r=1}^4 \mathcal{T}_{\text{AS},4}(r). \end{aligned}$$

All factors use the same ordering 1, 2, 3, 4. Thus  $\mathcal{A}_4$  is the full union of the four levels, not only the level-four component.

A stochastic matrix is *scrambling* if every two rows have a common positive column. For a class  $\mathcal{C}$ , its uniform scrambling horizon is

$$h_{\text{scr}}(\mathcal{C}) = \min\{h \geq 1 : P_h \cdots P_1 \text{ is scrambling for all } P_1, \dots, P_h \in \mathcal{C}\}.$$

Hsu proves the dimension-only bound  $h_{\text{scr}}(\mathcal{A}_4) \leq (4-1)(4-1)! = 18$  and records sharpness there only in order three [1, Theorem 5 and Remark 5]. The result below determines the order-four value within that framework. It is a comparison with the cited version of the preprint, not a claim of global priority.

**Theorem 1.** *For the full order-four almost-Sarymsakov class  $\mathcal{A}_4$  with one fixed common ordering,*

$$h_{\text{scr}}(\mathcal{A}_4) = 11.$$

*This equality holds uniformly for every assignment of positive stochastic weights having an admitted support.*

## 2 Reduction to Boolean supports

Represent a Boolean support by its four nonzero row masks  $r_i \in \{1, \dots, 15\}$ . If bit  $j-1$  of  $r_i$  is one, then entry  $(i, j)$  is supported. We pack the masks as the hexadecimal integer

$$m = r_1 + 2^4 r_2 + 2^8 r_3 + 2^{12} r_4.$$

Thus a displayed word **d4d3d2d1** has row masks **d1**, **d2**, **d3**, **d4**, in that order. Write  $A \odot B$  for Boolean matrix multiplication.

**Lemma 2** (Exact support reduction). *For nonnegative matrices  $P_1, \dots, P_h$ ,*

$$\text{supp}(P_h \cdots P_1) = \text{supp}(P_h) \odot \cdots \odot \text{supp}(P_1).$$

*Moreover, membership in  $\mathcal{A}_4$  depends only on the support. Every row-nonempty support has a positive row-stochastic realization.*

*Proof.* An entry of a nonnegative product is positive exactly when at least one positive path contributes to it; cancellation is impossible. This is the definition of the corresponding Boolean product. GE, GR, and all row-allowability requirements refer only to positive-entry sets, so the admission predicate also depends only on the support. For a row support  $S_i \neq \emptyset$ , assigning weight  $1/|S_i|$  to each supported entry in row  $i$  gives the asserted stochastic realization.  $\square$

The next elementary fact justifies pruning the exhaustive search as soon as a partial product becomes scrambling.

**Lemma 3** (Scrambling is left-absorbing). *If  $B$  is a scrambling Boolean support and  $A$  is row-allowable, then  $A \odot B$  is scrambling.*

*Proof.* Fix rows  $i, j$  of  $A \odot B$ . Choose a supported index  $k$  in row  $i$  of  $A$  and a supported index  $\ell$  in row  $j$ . Since  $B$  is scrambling, rows  $k$  and  $\ell$  of  $B$  meet in some column. That column occurs in both rows  $i$  and  $j$  of  $A \odot B$ .  $\square$

### 3 Complete enumeration and certificate

Let  $\mathcal{L}$  be the set of all Boolean supports admitted at at least one of the four levels. The enumeration visits each of the  $15^4 = 50,625$  row-nonempty supports once. For each support and each  $r \in [4]$ , it tests:

1. row-allowability and GR of the leading  $r$ -by- $r$  block;
2. row-allowability and GE of every leading  $s$ -by- $s$  block with  $r < s \leq 4$ ; and
3. a supported entry in columns  $1, \dots, r$  for every row below  $r$ .

GE and GR are checked for all ordered pairs of disjoint nonempty row subsets. Before deduplication, the numbers admitted at levels 1, 2, 3, 4 are

$$4096, \quad 15806, \quad 25462, \quad 35347.$$

Their Boolean union contains  $|\mathcal{L}| = 37,833$  supports.

Starting with the Boolean identity, define complete nonscrambling frontiers by

$$\begin{aligned} F_0 &= \{I\}, \\ F_{d+1} &= \{A \odot B : A \in \mathcal{L}, B \in F_d, A \odot B \text{ is nonscrambling}\}. \end{aligned} \tag{1}$$

**Proposition 4** (Frontier completeness).  *$F_d$  is exactly the set of distinct nonscrambling Boolean products of legal words of length  $d$ .*

*Proof.* The assertion is immediate for  $d = 0$ . Suppose it holds at depth  $d$ . Every state constructed by (1) is a nonscrambling legal product of length  $d + 1$ . Conversely, let  $A \odot B$  be a nonscrambling legal product of length  $d + 1$ , where  $A \in \mathcal{L}$  is its leftmost factor. If  $B$  were scrambling, Lemma 3 would make  $A \odot B$  scrambling. Thus  $B$  is nonscrambling, belongs to  $F_d$  by induction, and the product is retained by (1).  $\square$

The complete computed cardinalities are as follows.

| $d$     | 0 | 1     | 2     | 3    | 4    | 5    | 6    | 7   | 8   | 9   | 10  | 11 |
|---------|---|-------|-------|------|------|------|------|-----|-----|-----|-----|----|
| $ F_d $ | 1 | 19946 | 13532 | 9802 | 6258 | 3332 | 1720 | 988 | 566 | 316 | 120 | 0  |

In particular, Proposition 4 and  $F_{11} = \emptyset$  prove that every legal eleven-factor support product is scrambling.

This conclusion is certified state by state, rather than only by the counts in the table. The generator serializes the complete sorted legal alphabet and every complete sorted frontier. A separate checker scans all  $2^{16}$  Boolean relations, reconstructs GE and GR using ternary assignments of each row index to neither set, the first set, or the second set, and compares all 37,833 alphabet entries and 56,581 frontier entries with the serialized certificate. It also recomputes the witness below by an independent Boolean multiplication routine. Both implementations use integer bit operations only; there is no floating-point arithmetic or random sampling.

At the level of proof design, separating an exact finite-state model, the mathematical theorem, and independent exact computations used as checks also appears in a cellular-automaton period problem [2]. That work concerns a different discrete system and supplies no assertion about stochastic matrices, Sarymsakov classes, or scrambling; it is cited only as a methodological comparison.

## 4 A maximal nonscrambling word

The following ten legal supports give the matching lower bound. The third column lists one level that admits each factor, and the fourth gives the cumulative Boolean product  $P_t \odot \cdots \odot P_1$ .

| $t$ | $P_t$ | admitted level | cumulative product |
|-----|-------|----------------|--------------------|
| 0   | —     | —              | 8421               |
| 1   | a126  | 2              | a126               |
| 2   | 423a  | 3              | 126a               |
| 3   | c586  | 4              | 3a16               |
| 4   | a215  | 2              | 316e               |
| 5   | 4643  | 3              | 171e               |
| 6   | 21a6  | 2              | 1e17               |
| 7   | 42a3  | 3              | e117               |
| 8   | 9368  | 4              | f71e               |
| 9   | 3126  | 2              | fe17               |
| 10  | 2345  | 3              | 17ef               |

The final support 17ef has row masks  $[\mathbf{f}, \mathbf{e}, 7, 1]$ . Its second and fourth rows are disjoint because  $\mathbf{e} \& 1 = 0$ ; hence this legal ten-factor product is not scrambling. Lemma 2 turns every factor into an explicit stochastic matrix, for example by putting equal mass on its supported entries. Thus  $h_{\text{scr}}(\mathcal{A}_4) > 10$ . Together with the empty eleven-step frontier, this proves Theorem 1.

## Reproducibility and limitations

The ancillary files `search_exact.cpp` and `verify_certificate.cpp` respectively generate and independently check `frontier_certificate.txt`. The human-readable outputs record the alphabet size, every frontier count, the horizon, and the witness product. The certificate is finite and exact, but it concerns precisely the fixed-order full class defined above. It does not cover a convention in which factors may use independently permuted orderings. The literature comparison was bounded; no claim of worldwide priority is made.

## Computational and AI assistance disclosure

The exhaustive enumeration and certificate verification were performed by two C++ programs using exact Boolean operations. A generative AI system assisted with proof organization and manuscript drafting. The finite certificate, support translation, and mathematical argument were checked in a separate audit; AI-generated text was not treated as mathematical evidence.

## References

- [1] Shun-Pin Hsu. Sarymsakov-type semigroups with support dominance and nested mixing cores. arXiv preprint arXiv:2609.05050v1, 2026. <https://arxiv.org/abs/2609.05050>.
- [2] Zijian Zeng. Exact odd-parameter periods for reversible elementary second-order cellular automaton rule 115. SSRN preprint, 2026. doi: 10.2139/ssrn.7379379.